Subjective, But Not Worthless
- Non-linguistic Features of Chatterbot Evaluations

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Abstract

This paper is a contribution to the heavily neglected field of chatterbot evaluation methods. Based on our experience from research on humor-equipped chatterbots, we propose a methodology for dialogue system evaluation, focusing on their non-task-oriented subtype. Using examples from our previous experiments, we discuss several evaluation methods, discuss what to evaluate, how to do that and how to analyze the results.

1. Introduction

Nowadays, there is a growing need for spoken language dialogue systems (SLDS). To address that need, many research projects are being launched and the field of freely talking computers is gaining more and more interest.

There are two commonly known types of conversational systems: task oriented (such as virtual kiosks or tour guides) and non-task oriented ("chatterbots"). As the area is relatively new, there is an urgent need to work out a robust methodology, also in the field of evaluation. Especially for research on chatterbots the methodology is still quite neglected and there is a lack of even basic standardization.

1.1 Two Areas of Evaluation

Evaluation of dialogue systems depends strongly on their purposes and design. Most existing research projects focus on areas which can generally be divided into two groups:

1) focused on linguistic skills and/or technical quality;

2) focused on non-linguistic skills.

The first area is basically common for both task- and non-task oriented systems. It concerns the system's linguistic skills, such as grammatical correctness, semantic naturalness or vocabulary richness, as well as the technical quality of interaction (response time, voice recognition and generation, visual quality etc.). This area of evaluation is relatively objective.

The second area of evaluation differs for task- and non-task oriented systems. Systems belonging to the former type are designed to achieve specified goals, and this, in most cases, can be seen as the priority in the non-linguistic skills focused aspect of the evaluation. In our opinion, this presence of a specified goal, mutual for user and computer, makes evaluation in this area somehow easier to conduct, as such criteria can be easily and relatively objectively verified.

In the case of chatterbots, however, the non-linguistic skills focused area of evaluation cannot be as easily defined, as there is no mutual goal of the conversation. It is the pleasure of having the interaction that counts. In other words, evaluation of such systems must focus on the user's impressions of the features of the interaction that make it more pleasant, natural and generally "better" in the eyes of the user. Thus, by definition, such assessment has to be subjective.

However, this subjectivity does not necessarily have to be a drawback in chatterbot evaluation. In the end, this is what we want to check – the user's subjective opinion of the product. Another question is how we check it, or – which exact features of the interaction are worth investigating in order to give us the desired results.

1.2 Methodological Gap

Although there are some work reviewing existing methods used to evaluate task-oriented systems (e.g. Dybkjær and Bernsen, 2004), to our knowledge, no robust evaluation methodology for non-task oriented systems has been proposed so far.

In fact, this is the reason why we do not directly compare our methods with other existing research – there is simply nothing we could compare to (we do, however, discuss some particular methods – see below).

In this paper we focus on the non-linguistic area of chatterbot evaluation. However, the methods described here can also be applied in experiments on task-oriented systems, with some slight changes in their content.

2. Systems Used in This Research

The methods and perspectives of chatterbot evaluation presented in this paper have been used in our research on humor-equipped talking systems. In this section we briefly describe two chatterbots used in the research along with an emotiveness analysis system, used in the automatic evaluation experiments (see **4.3**).

2.1 Two chatterbots

The baseline system in our research is a Japanese, text based chatterbot called "Modalin" (developed by Higuchi et al, 2008). The system extracts keywords from user utterances, uses them to extract word associations from the Internet and adds modality to the generated responses (for details, see Higuchi et al. 2008).

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The chatterbot was used as a base to create a Japanese pun-telling system "Pundalin". To do that, we added a pun generating engine developed in our previous research (Dybala et al., 2008) to Modalin, using a very simple timing rule - in every third turn of the conversation, the system's output was replaced by a joke-including sentence, generated by the joking system (for details, see Dybala et al., 2008). Currently we are working on a more sophisticated algorithm.

2.2 ML-Ask System

In our research we used the ML-Ask Emotive Elements/Emotive Expressions Analysis System for Japanese (Ptaszynski et al., 2008) to perform automatic analysis of chat logs acquired in the user-focused experiment (see **4.3**). The ML-Ask system performs utterance analysis in two general steps:

1. Determining general emotiveness (emotive/non-emotive), and
2. Specifying valence and types of emotions found (positive/negative plus specified type).

In the second step, analysis of the specific emotions showed by the evaluators during the conversation provided us with the information of their feelings towards the system.

In our research, we separated the emotion types and emotion expressions using two general dimensions: positive/negative and activated/deactivated (Russel, 1980). Each type of emotion can be described using these two dimensions.

The types of emotions in our research are based on Nakamura's (1993) Japanese emotion classification (10 types). The proposed emotion types were projected on Russel's 2-dimensional model of affect (Russel, 1980). The effect of this projection can be seen in Figure 1.

3. Our Evaluation Methodology

The subjectivity of non-linguistic features of chatterbots evaluation and the need for "measuring" user impressions of the system do not give us much of a choice when conducting experiments. The easiest and most obvious method to do that is to ask users directly what they think of the interaction.

Although of high importance, evaluation conducted by users also has its drawbacks (discussed below). Thus, in our research we employed two complementary, non-user focused evaluation methods: third person focused evaluation and automatic (emotiveness analysis based) evaluation.

Figure 2 presents the outline of the methodology proposed in this paper. Needless to say, it still needs to be improved – we hope this paper will trigger a thorough discussion on this subject.

Below we describe and discuss the methods from the non-linguistic area, using examples from our earlier research. In this paper we focus on non-task oriented systems.

4. What to Evaluate and How?

This may sound completely obvious, but we need to state one thing before we start explaining and discussing our evaluation methods: the shape of the evaluation experiment strongly depends on what we want to check. In our research we focus on non-task oriented humor-equipped dialogue systems. In the evaluation experiment we explore both the linguistic- and non-linguistic area – however, we focus on the latter, as it is the role of humor in the interaction we want to check in the first place. For the same reason, some of the questions in our evaluation were directly related to humor and perceived funniness of the systems' utterances.

Taking these two aspects into consideration, in our evaluation we decided to explore such non-linguistic aspects of interaction as: human-likeness, the will to continue the dialogue, engagement in the conversation, funniness and emotive response. We also evaluated the linguistic area of the systems' performance – however, in our research these results were of secondary importance.

These issues are described in details below.

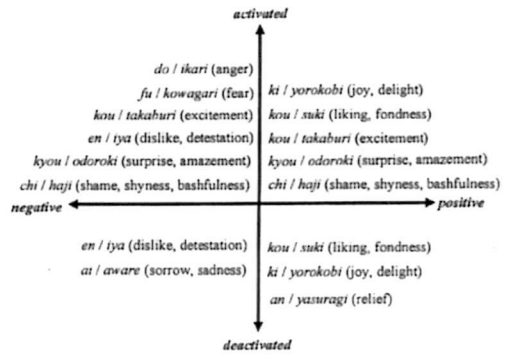


Figure 1 Grouping Nakamura's classification of types of emotions on Russell's two-dimensional space

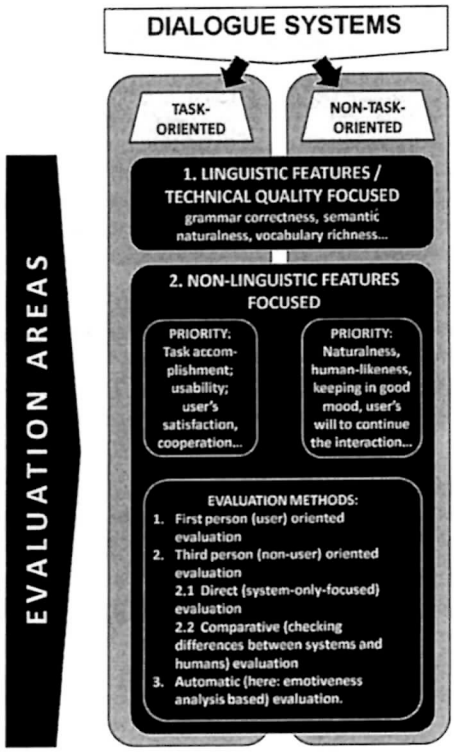


Figure 2 Evaluation areas of task- and non-task oriented dialogue systems

4.1 First Person Oriented Evaluation

As it is users who are the "clients" of our product, in the research on dialogue systems the first person oriented evaluation is to us of highest importance. Although not perfect, methods described in this section allow us to check the user impressions of the interaction with the system, in possibly the most direct way. As stated above, this evaluation is by definition subjective, but we do not see it as its drawback. Instead, we propose to accept the lack of objectivity as a natural fact in impression-relying evaluation. Individual differences are inevitable here, and their presence can be a trigger to construct more sophisticated, user-adapting systems. An idea of such a system for humor-oriented chatterbots is described in one of our earlier works (Dybala et al., 2009).

There are two major methods of first person oriented evaluation: one in which user impressions of the interaction are checked by filling out a questionnaire or conducting an interview, and another one conducted using the Turing Test (Turing, 1950). The latter has for years been the subject of many discussions and actually received a lot of criticism. Therefore, as this method is probably the best known of all methods mentioned in this paper, we are not going to discuss it in detail here. We would only like to mention that – comparing to the questionnaire/interview method – the Turing Test does not give us detailed and measurable information about the interaction, since its only aim is to check if the system is sophisticated enough to deceive users about it not being human.

Therefore, in our research we used the former first person evaluation method to conduct the first person oriented evaluation experiment. Human evaluators were asked to perform a 10-turn dialogue with Modalin (non-humor-equipped), and then with Pundalin (humor-equipped system). No topic restrictions were made.

There were 13 participants in the experiment, 11 male and 2 female; all of them were university undergraduate students. After talking with both systems, they were asked to fill out a questionnaire about each system's performance. The questions concerned both linguistic (B-D) and non-linguistic (A, E-H) areas of interaction:

A) Do you want to continue the dialogue with the system?; **B)** Was the system's output grammatically natural?; **C)** Was the system's output semantically natural?; **D)** Was the system's vocabulary rich?; **E)** Did you get an impression that the system possesses any knowledge?; **F)** Did you get an impression that the system was human-like?; **G)** Do you think the system tried to make the dialogue more interesting?; **H)** Did you find the conversation with the system interesting?

The replies to the questions were given on 5-point scales with some explanations added. Each evaluator filled out two such questionnaires, one for each system. The final, summarizing question was "Which system do you think was better?". Statistical significance of the results was calculated using the Student's t-test. The results are summarized in Table 1.

Question	Modalin	Pundalin	Difference	P value
A	2.62	3.38	**0.76**	>0.05
B	2.15	2.92	**0.77**	>0.05
C	1.85	2.69	**0.84**	**<0.05**
D	2.08	3.00	**0.92**	**<0.05**
E	2.15	2.85	**0.70**	**<0.05**
F	2.38	3.31	**0.93**	**<0.05**
G	1.92	4.15	**2.23**	**<0.05**
H	2.46	4.08	**1.62**	**<0.05**
Which is better?	15%	85%		

Table 1 User evaluation – results for Modalin and Pundalin for detailed questions. Answers were given on a 5-point scale.

The results show that the system with humor received higher scores in both linguistic and non-linguistic areas. As for the former, it may seem unusual that the presence of humor improved the system's linguistic skills – this fact, however, could have been caused by the fact that Pundalin uses fragments of human created sentences and jokes from a data base, which naturally are more correct than those automatically generated by the computer.

Also in the non-linguistic area all results point at the humor-equipped system. Users wanted to continue the conversation to a higher degree with Pundalin than with Modalin, perceived Pundalin as more human-like, knowledgeable, funny and generally better than Modalin (Dybala et al., 2008).

Results for questions A and B were found to be significant on the 6% level, and for remaining questions – on the 5% level.

Discussion

As mentioned above, the first person oriented method is the best and most direct way of evaluating the system. Such a method (questionnaire/interview) was also used by Bernsen and Dybkjær in their experiments on NICE - a system for spoken and gesture interaction with life-like fairytale author Hans Christian Andersen (Bernsen and Dybkjær, 2004). The content of the questions was slightly different from those we used, as the embodiment of the system and the usage of gestures also needed to be addressed. However, the general tendency in NICE's evaluation was consistent with ours: most questions concerned the users' subjective impressions of the interaction (such as: "How did it feel to talk to the system?"). The biggest difference between our and Bernsen and Dybkjær's evaluation was qualitative – e. g., the answers in their experiment were given freely by the users, without any quantitative scale. User responses were manually analyzed and their descriptive summarization is the result of the experiment. Such non-quantitative methods surely can give us a deeper insight into the users' impressions of the system – however, it is quite hard to use when comparing with other systems. Therefore, we think that the qualitative methods should be used in preliminary experiments, in order to receive feedback from the users, rather than to evaluate the final product.

While of high importance, user-focused evaluation also has its drawbacks. First, even when conducted immediately after the interaction, it requires the user to remember

his/her impressions from when the conversation took place. On the other hand, the alternative here would be asking the user to evaluate the system during the conversation, which might distract him/her and negatively influence the smoothness of the interaction. One way to solve this problem is to also conduct a third person oriented experiment (see **4.2**).

Another problematic issue of the first person oriented evaluation method is that we cannot be sure if the user is actually aware of his/her own feelings and emotions. In our research we partially solved this problem by using the automatic emotiveness analysis based evaluation. Even if the users do not fully realize their feelings and do not reflect them in the evaluation, there is a chance that they will be reflected in what they say. Therefore, emotiveness analysis of the user utterances can be seen as a good complementary evaluation method.

4.2 Third Person Oriented Evaluation

In the first person oriented evaluation we ask the users directly about their impressions of the system. However, as mentioned above, one drawback of this method is that the evaluation has to be conducted after the conversation. To solve this problem and double-check the results of the first experiment, we conducted an additional experiment, in which third person (non-user) participants evaluated the chat logs from the user experiment. The questions asked were similar to those used in the user-focused experiment – we only made minor adjustments. First, the word "system" was changed to "dialogue" (in some cases - "Speaker"), as we did not want the evaluators to know that some of the utterances were generated by a computer system. In the chat logs given to the third person evaluators, dialogue participants were called "Speaker A" for the user and "Speaker B" for the system. In addition, question F (about human-likeness) was deleted, as it would also reveal that at least one speaker was not human. Also, in questions B, C, D, E, G and H we added two options: 1) "Speaker A" and 2) "Speaker B" – so that the dialogue participants would be evaluated separately. Thus, the list of questions used in this experiment goes as follows:

A) Do you want to read the continuation of the dialogue?; **B)** Was Speaker A/B's talk grammatically natural?; **C)** Was Speaker A/B's talk semantically natural?; **D)** Was Speaker A/B's talk vocabulary rich?; **E)** Did you get an impression that Speaker A/B possesses any knowledge?; **F)** <deleted>; **G)** Do you think the Speaker A/B tried to make the dialogue more funny and interesting?; **H-1)** Did you find the dialogue interesting in general? **H-2)** Did you find Speaker A/B's talk interesting?

After completing the detailed questionnaire, the evaluators answered the final question, the same as in the previous experiment - "Which dialogue did you find most interesting?" (we used the Japanese word *omoshiroi*, which can mean "interesting" or "funny", and is generally positive in meaning (Nakamura, 1993)).

The chat logs were divided into 13 sets. Each of them included one Modalin and one Pundalin dialogue. Each set was evaluated by 5 participants, which makes a total of 65 evaluators, all of which were university students. (Dybala et al., 2008). Statistical significance of the results was calculated using the Student's t-test.

The results were analyzed using two methods mentioned above: direct and comparative.

4.2.1 Direct Evaluation

In this method we only take into consideration the results for systems' (Speaker B's) utterances, and compare them for both humor- and non-humor equipped system, as we did in first person evaluation. The results of this method are summarized in Table 2.

Although the differences here were not that clear and significant as in the user-focused evaluation, the tendency is still visible. The humor-equipped system received higher scores in all categories. Only in two cases (D and F) were the differences found to be statistically significant – however, the results for the general question show that even if there is not much difference, the evaluators still chose dialogues with humor (69% vs. 31%).

Question	Modalin	Pundalin	Difference	P value
A	2.60	2.89	**0.29**	>0.05
B	1.78	2.09	**0.31**	>0.05
C	1.48	1.69	**0.21**	>0.05
D	2.03	2.38	**0.35**	**<0.05**
E	1.87	2.13	**0.26**	>0.05
F	X	X	X	X
G	2.51	2.91	**0.40**	**<0.05**
H-1	2.88	3.19	**0.31**	>0.05
H-2	2.73	3.16	**0.43**	>0.05
Which is better?	31%	69%		

Table 2 Third person evaluation – results for Modalin (non-humor equipped system) and Pundalin (humor-equipped system). Answers were given on a 5-point scale.

Discussion

As shown in the Table 2, the results are generally consistent with those of the first person oriented experiment. Thus, it can be stated that the direct third person oriented method can be used to evaluate chatterbot performance. However, the low differences and lack of significance in this experiment require some detailed discussion.

One possible explanation of this phenomenon is that when users are talking with the systems, they are usually quite impressed by the very fact that a computer can talk. This very fact may positively influence the results. With this understanding, the third person oriented evaluation seems more objective, since the evaluators were not participants of the interaction, and thus had more distance to the subject of evaluation. Also the fact that they did not know that one of the speakers was a computer system was not without meaning.

Albeit the relative "objectiveness" of the third person evaluation (more distance towards chat logs than towards conversation partner), this method has a few drawbacks. The major one is that, as mentioned above, it is the user that has to be satisfied in the first place. They will use the

system and it is their opinion that counts the most. Of course, the more severe and diversified the evaluation, the more information about our systems and enhancements needed we can get; however, to evaluate the final product it is still the first person oriented evaluation that should be of primary importance, and third person oriented methods should be used rather to double-check the results or to acquire feedback leading to the system's development.

4.2.2 Comparative Evaluation

As mentioned above, in our third person evaluation experiment we referred to the speakers in the chat logs as Speaker A (the user) and Speaker B (the system). In the direct evaluation we took into consideration only the results for Speaker B, while in the comparative evaluation we calculated the differences between the systems and the users. Statistical significance of all scores was calculated using the Student's t-test. The results are summarized in Tables 3 and 4.

Modalin				
Question	**User**	**System**	**Diff.**	**P value**
B	3.30	1.78	**1.52**	<0.05
C	2.94	1.48	**1.46**	<0.05
D	2.92	2.03	**0.89**	<0.05
E	3.13	1.87	**1.26**	<0.05
G	2.54	2.51	**0.03**	>0.05
H-2	2.85	2.73	**0.12**	>0.05

Table 3 Results for Modalin for detailed questions in third person evaluation (differences between users and systems). Negative values mean that Speaker B (the system) received higher scores than the user.

Pundalin				
Question	**User**	**System**	**Diff.**	**P value**
B	3.18	2.09	**1.09**	<0.05
C	3.00	1.69	**1.31**	<0.05
D	2.81	2.38	**0.43**	<0.05
E	2.97	2.13	**0.84**	<0.05
G	2.52	2.91	**-0.39**	<0.05
H-2	3.09	3.16	**-0.07**	>0.05

Table 4 Results for Pundalin for detailed questions in third person evaluation (differences between users and systems). Negative values mean that Speaker B (the system) received higher scores than the user.

As shown in the above tables, the results show that the humor-equipped system differs less from humans than the non-humorous one. In other words, the difference between humans and Pundalin was smaller than the difference between humans and Modalin. In our research this is especially important for questions D-H, which belong to the non-linguistic area of evaluation. Looking at the results, we can see that the system with humor actually made more effort than the users to make the dialogue interesting. The fact that Pundalin surpassed the users in this category can be interpreted as not necessarily positive, as trying too hard may also be annoying. However, knowing that the system was assessed as generally better both by the users and third person evaluators, we can assume that the attempts to make the conversation more interesting were rather appreciated than disliked.

Discussion

From these results, a conclusion can be drawn that the system which differs less from humans can be seen as more human like. This assumption is consistent with the results of the first person oriented experiment (see **4.1**, question F). Obviously, more research on the issue of human-likeness is needed – however, we think that the method suggested here is also an option for checking how close to the human level the system is.

This method, albeit innovative, has also several drawbacks. The main one is the same as in case of the direct evaluation (see **4.2.1**) – it may be slightly less subjective, but it is also less direct, and does not involve the users. However, the consistency with the results of the user oriented experiment shows that it can be used as a complementary method of evaluation.

In previous sections we mentioned the Turing Test, as probably the best known (although arguable) method for checking the system's human likeness. The Turing test is a first person oriented method, in which the users have to tell if the interlocutor is a human or a computer. However, it should be possible to conduct a third person oriented version, in which the evaluators would read the chat logs and guess the identity of speakers. Obviously, third person oriented Turing Test would have the same drawback as other third person oriented methods – however, we believe that it can be a good complementary method and as such may be worth trying.

4.3 Automatic Evaluation

In the above sections we discussed some drawbacks of first and third person oriented evaluation methods (see **4.1** and **4.2**). These problems can be solved by using automatic methods, in which the system's performance is evaluated by another system. In our research we used the ML-Ask System (see **2.2**) to perform automatic analysis of chat logs acquired in the user-focused experiment.

4.3.1 General Emotiveness

In the first step, general emotiveness (emotive/non-emotive) of user utterances was analyzed by the ML-Ask system. Most of the users (11 out of 13) showed more emotions towards Pundalin than towards Modalin, which means that they were generally more emotively engaged in the conversation with the system which used humor (Ptaszynski et al., 2008).

4.3.2 Valence and specification

In the next step of the evaluation, the chat logs were analyzed to check the specific types of emotions in the users' utterances. Figure 4 shows the results projected on the Russel's two-dimensional space (see also **2.3**, Figure 1).

As shown above, while most of emotions towards Modalin were negative and activated (45%, 78% of negative emotions in total), for Pundalin the proportion

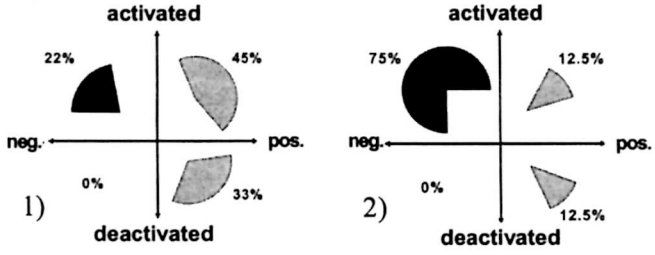


Figure 3: Projection of emotive analysis of users' emotions types on Russell's two-dimensional space – 1) Modalin (without humor) and 2) Pundalin (with humor)

was opposite (75% of positive and activated). In this experiment, no positive deactivated emotions were found in either the humor- nor non-humor-equipped system's chat logs.

4.3.3 Positive/Negative Engagement

The correlation between the speaker's emotiveness and conversation engagement has been shown in various research (e.g. Goodwin and Goodwin, 2000). This knowledge was also used by Yu et al. (2004) in their research, in which they measured engagement level based on emotions recognized in user speech. The approach is quite similar to the one presented in this section – however, in our research, we used emotions extracted from the textual layer of the conversation, and focused on the influence of humor as a measure to improve the engagement.

Conclusions about the users' engagement in the conversation were drawn based on the results described in **4.3.1** and **4.3.2**. General emotiveness analysis results (see Figure 3) suggest that the users were more engaged in the conversation with Pundalin, as they showed more emotions towards it than towards Modalin. However, the engagement does not necessarily have to be positive, as the users might have reacted to the systems' utterances with negative arousal or irritation. This is why we propose to distinguish between positive and negative engagement. To do that, in our research we use emotive valence (positive/negative emotions – see Figure 3). The results show that the users were generally more positively and less negatively engaged in the conversation with Pundalin. This is consistent with the results of the previous experiments, especially with the questions concerning the will to continue the interaction (see **4.1**). Therefore, it can be stated that automatic emotiveness analysis can also be used to investigate user engagement and its types.

The role of activation/deactivation of emotions is still to be investigated.

Discussion

In this section we described another automatic evaluation method used in our research. This method has a few significant advantages. First: it is automatic. This means that it does not require any additional engagement from the users in the experiment – they only have to perform a conversation with the system. We do not have to waste our or other people's time. The evaluation can be conducted at any time, not necessarily right after the conversation.

Second: as mentioned above, it is quite difficult to speak about one's own emotions and feelings, as we may not be fully aware of all of them. Not mentioning the fact that – if the evaluation is conducted after the conversation – users may not exactly remember what they felt some time ago. These feelings and emotions, however, can be revealed in the users behaviour during the interaction, and also in the textual layer, which, in case of our text-based-only system, is of high importance.

Of course, even during conversation with text-based systems like ours, emotions can also be detected from other layers than the textual. Such methods as voice or facial expression recognition-based emotions detection should also be taken into consideration (and, possibly, combined with the textual-based one we used).

5. Conclusions

In this paper we proposed a methodology for chatterbot evaluation, focusing on the non-linguistic area. The proposed methods have been experimentally tested, so it can be stated that they are at least applicable and can be used also in other research on non-task oriented dialogue systems.

Obviously, the methods and methodology are not perfect and there are many more issues that have to be taken into consideration, such as system embodiment, voice recognition/generation or user-system interaction length and frequency.

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